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Milestone Three Narrative

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The artifact I selected for the algorithms and data structure category is my ABCU Advising Assistance Program from CS 300. This program was originally created as the final project in that course and was designed to load a catalog of university courses from a CSV file, store them in memory, and allow users to view either the entire course list or the details of a single course, including its prerequisites. The original version used a simple vector-based structure with linear searching, minimal input handling, and basic menu interactions. For the purposes of my CS 499 enhancement, I intentionally chose this artifact because it focuses directly on algorithmic thinking and the choice of data structures, making it a strong representation of the outcomes related to efficiency, organization, and computational problem solving.

I included this artifact in my ePortfolio because it clearly demonstrates both my understanding and my growth in the areas of algorithms and data structures. In its enhanced form, the artifact showcases several improvements that directly align with advanced algorithmic principles. The most significant enhancement was restructuring the core of the program to use an unordered_map instead of a vector. This change transformed course lookups from linear time to average constant time, immediately improving both performance and design quality. I also created algorithms for input normalization, which included trimming whitespace and converting all user queries to uppercase. Additionally, I implemented a sorting algorithm for printing course lists in consistent, alphabetical order using a vector of extracted keys and standard library sorting. Another improvement was the addition of a suggestion system that scans course numbers and titles for partial matches when the user enters an incorrect or incomplete course identifier. This feature demonstrates algorithmic reasoning, iteration over stored data, and the use of string-matching logic. Finally, I improved the file-parsing algorithm by adding validation, detecting malformed lines, and using a temporary data structure to avoid corrupting the catalog in the event of a parsing error. Together, these enhancements strengthen the artifact in a way that directly aligns with the program outcomes for algorithms, data structures, and reliable computational design.

In Module One, I planned for this artifact to demonstrate the program outcome related to designing and evaluating computing solutions using algorithmic principles and appropriate data structures. After completing the enhancement, I believe I fully met that planned outcome. The changes I implemented required clear reasoning about efficiency, organization of data, and how to produce a cleaner and more scalable solution. I do not need to update my outcome coverage plan, because this artifact continues to best represent my work in the algorithms and data structures category. I also believe it indirectly supports the communication outcome, because the improved error messages, suggestion prompts, and more structured outputs help demonstrate my ability to design software that communicates clearly with its users.

Enhancing this artifact taught me several important lessons. Revisiting earlier work with a more advanced skillset allowed me to recognize design limitations I could not fully address as a newer student in CS 300. One of the biggest lessons was how much clarity and efficiency can be gained simply by choosing the right data structure at the beginning. Using an unordered_map not only improved performance but also made the logic of the program easier to follow. I also learned the importance of input validation and normalization. Small details, such as trimming whitespace or converting user input to a standard form, prevent a wide range of errors and make the program more robust. Developing the suggestion feature helped reinforce my understanding of iterative search patterns, condition checking, string operations, and sorting. The biggest challenges I faced involved balancing program complexity with readability. For example, adding features like filename normalization and defensive file loading required careful structuring so the code remained clear and maintainable. Overall, this enhancement process strengthened my understanding of algorithmic design and validated the importance of thoughtful planning in software development. This improved version of the artifact better reflects my current abilities and is a strong representation of my readiness for professional work in the field of computer science.